

Signals & System

Lecture 4

Linear Systems and Convolution

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Lecture 4: Linear Systems and Convolution

Linear systems, Convolution: Impulse response, input signals as continuum of impulses. Convolution, discrete-time and continuous-time. LTI systems and convolution

Specific objectives for today:

We're looking at **discrete time** signals and systems

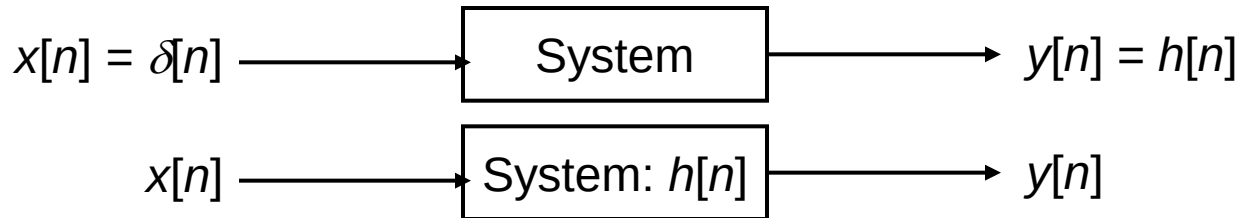
- Understand a system's **impulse response** properties
- Show how any input signal can be decomposed into a **continuum of impulses**
- **DT Convolution** for time varying and time invariant systems

Lecture 4: Resources

SaS, O&W, C2.1

Introduction to Convolution

Definition Convolution is an operator that takes an input signal and returns an output signal, based on knowledge about the system's unit impulse response $h[n]$.



The basic idea behind convolution is to use the system's response to a simple input signal to calculate the response to more complex signals

This is possible for LTI systems because they possess the **superposition** property (lecture 3):

$$x[n] = \sum_k a_k x_k[n] = a_1 x_1[n] + a_2 x_2[n] + a_3 x_3[n] + \dots$$

$$y[n] = \sum_k a_k y_k[n] = a_1 y_1[n] + a_2 y_2[n] + a_3 y_3[n] + \dots$$

Discrete Impulses & Time Shifts

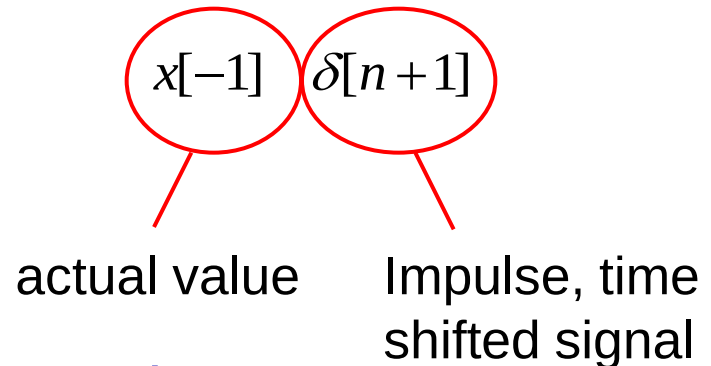
Basic idea: use a (infinite) set of discrete time impulses to represent any signal.

Consider any discrete input signal $x[n]$. This can be written as the linear sum of a set of unit impulse signals:

$$x[-1]\delta[n+1] = \begin{cases} x[-1] & n = -1 \\ 0 & n \neq -1 \end{cases}$$

$$x[0]\delta[n] = \begin{cases} x[0] & n = 0 \\ 0 & n \neq 0 \end{cases}$$

$$x[1]\delta[n-1] = \begin{cases} x[1] & n = 1 \\ 0 & n \neq 1 \end{cases}$$



Therefore, the signal can be expressed as:

$$x[n] = \cdots + x[-2]\delta[n+2] + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \cdots$$

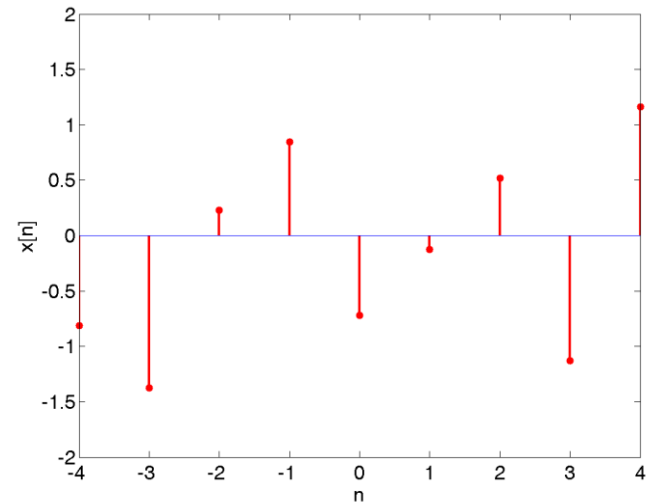
In general, any discrete signal can be represented as:

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

The **sifting property**

Example

The discrete signal $x[n]$



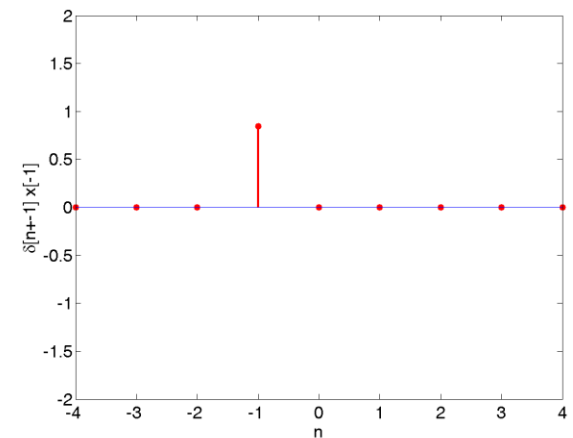
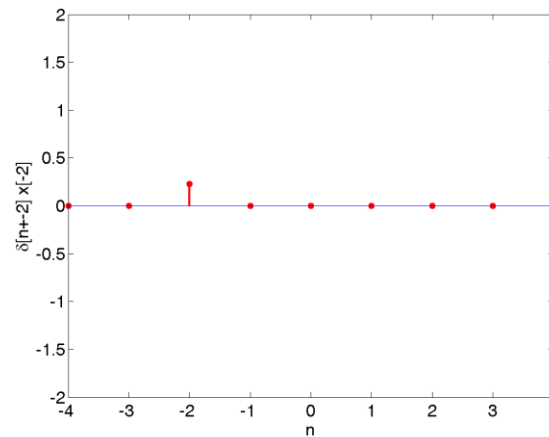
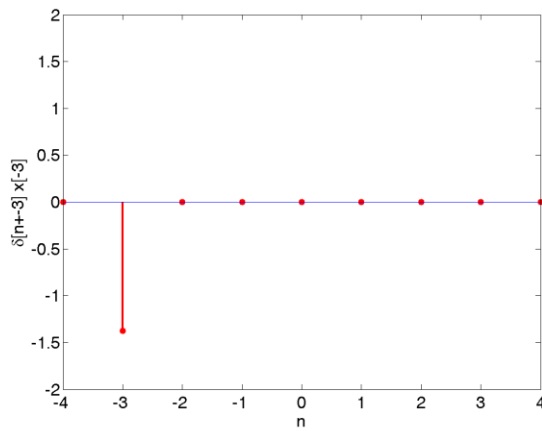
Is decomposed into the following additive components

$$x[-4]\delta[n+4] +$$

$$x[-3]\delta[n+3] +$$

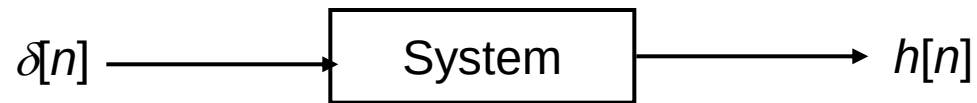
$$x[-2]\delta[n+2] +$$

$$x[-1]\delta[n+1] + \dots$$



Discrete, Unit Impulse System Response

A very important way to analyse a system is to study the output signal when a unit impulse signal is used as an input



Loosely speaking, this corresponds to giving the system a kick at $n=0$, and then seeing what happens

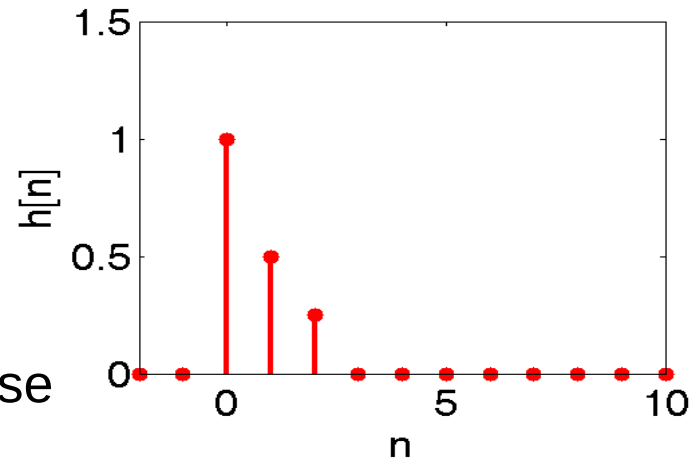
This is so common, a specific notation, $h[n]$, is used to denote the output signal, rather than the more general $y[n]$.

The output signal can be used to infer properties about the system's structure and its parameters.

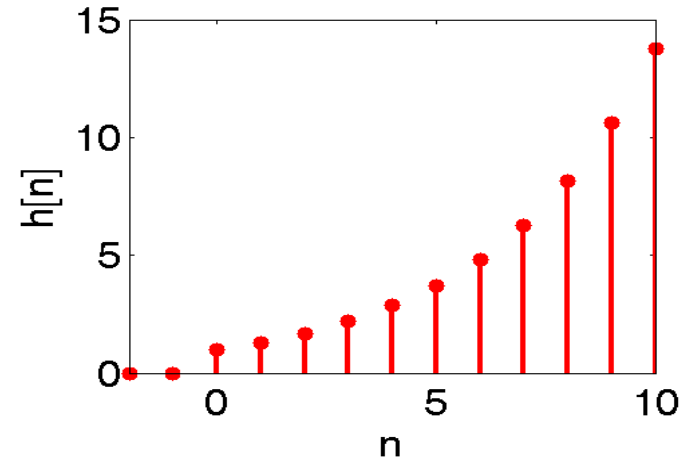
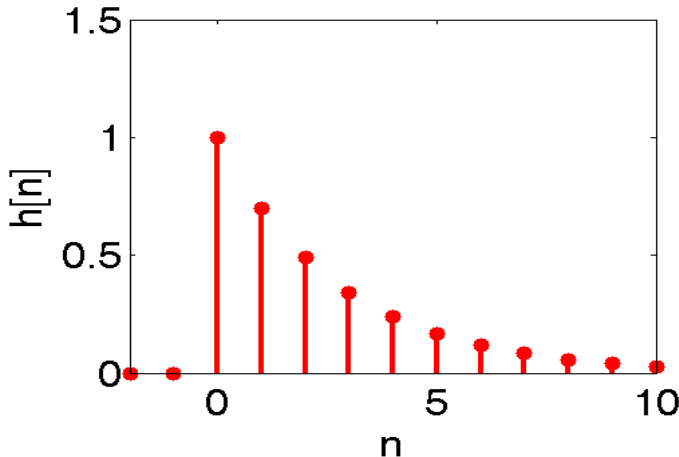
Types of Unit Impulse Response

Looking at unit impulse responses, allows you to determine certain system properties

Causal, stable, finite impulse response
 $y[n] = x[n] + 0.5x[n-1] + 0.25x[n-2]$



Causal, stable, infinite impulse response
 $y[n] = x[n] + 0.7y[n-1]$



Causal, unstable, infinite impulse response
 $y[n] = x[n] + 1.3y[n-1]$

Linear, Time Varying Systems

If the system is time varying, let $h_k[n]$ denote the response to the impulse signal $\delta[n-k]$ (because it is time varying, the impulse responses at different times will change).

Then from the superposition property (Lecture 3) of linear systems, the system's response to a more general input signal $x[n]$ can be written as:

Input signal

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$$

System output signal is given by the **convolution sum**

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h_k[n]$$

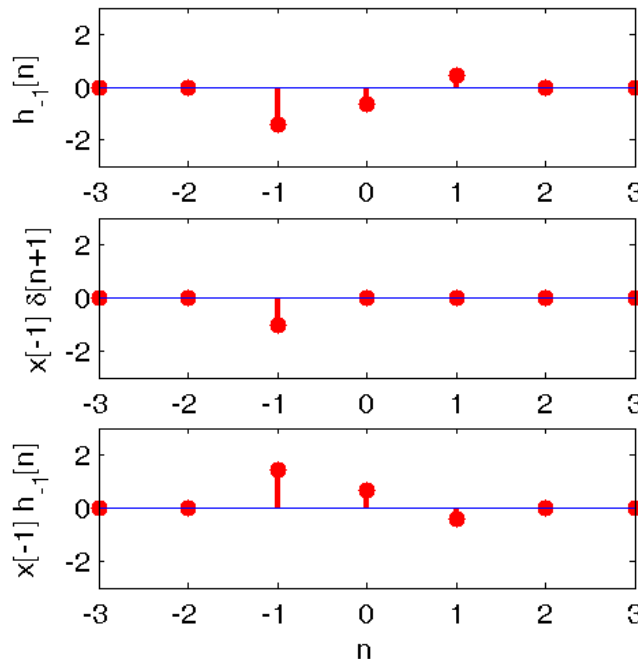
i.e. it is the scaled sum of impulse responses

Example: Time Varying Convolution

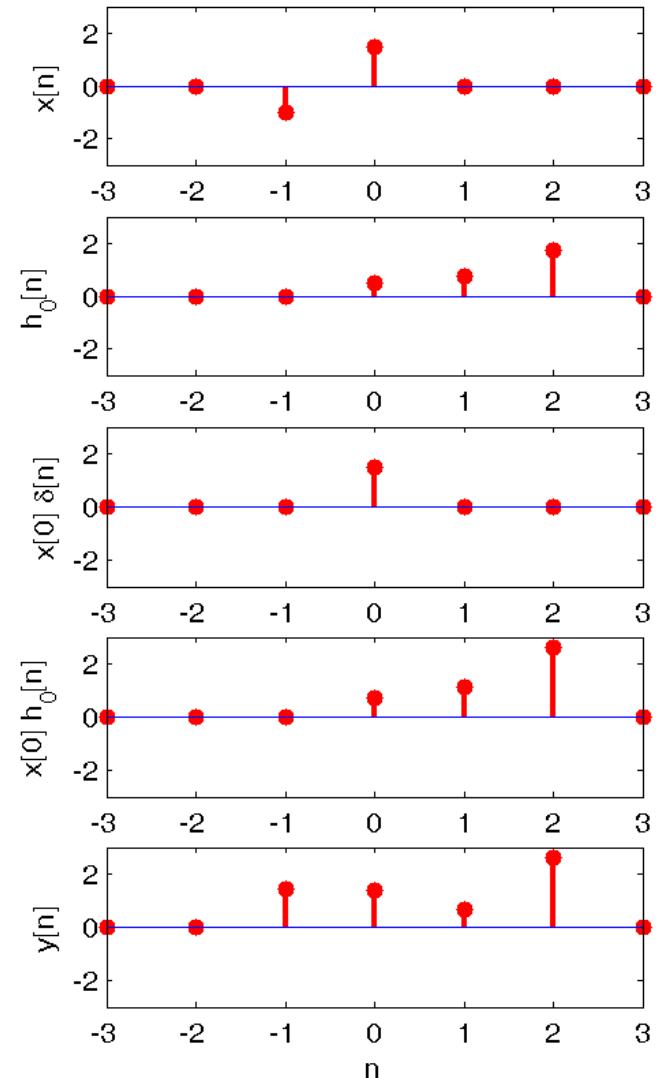
$$x[n] = [0 \ 0 \ -1 \ 1.5 \ 0 \ 0 \ 0]$$

$$h_{-1}[n] = [0 \ 0 \ -1.5 \ -0.7 \ .4 \ 0 \ 0]$$

$$h_0[n] = [0 \ 0 \ 0 \ 0.5 \ 0.8 \ 1.7 \ 0]$$



$$y[n] = [0 \ 0 \ 1.4 \ 1.4 \ 0.7 \ 2.6 \ 0]$$



Linear Time Invariant Systems

When system is linear, **time invariant**, the unit impulse responses are all time-shifted versions of each other:

$$h_k[n] = h_0[n - k]$$

It is usual to drop the 0 subscript and simply define the unit impulse response $h[n]$ as:

$$h[n] = h_0[n]$$

In this case, the convolution sum for LTI systems is:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n - k]$$

It is called the convolution sum (or superposition sum) because it involves the convolution of two signals $x[n]$ and $h[n]$, and is sometimes written as:

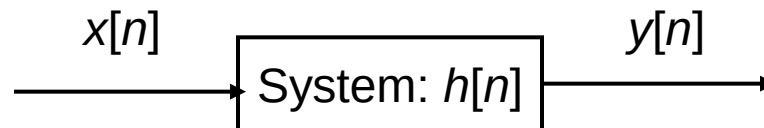
$$y[n] = x[n] * h[n]$$

System Identification and Prediction

Note that the system's response to an arbitrary input signal is completely determined by its response to the unit impulse.

Therefore, if we need to **identify** a particular LTI system, we can apply a unit impulse signal and measure the system's response.

That data can then be used to **predict** the system's response to any input signal



Note that describing an LTI system using $h[n]$, is equivalent to a description using a difference equation. There is a direct mapping between $h[n]$ and the parameters/order of a difference equation such as:

$$y[n] = x[n] + 0.5x[n-1] + 0.25x[n-2]$$

Example 1: LTI Convolution

Consider a LTI system with the following unit impulse response:

$$h[n] = [0 \ 0 \ 1 \ 1 \ 1 \ 0 \ 0]$$

For the input sequence:

$$x[n] = [0 \ 0 \ 0.5 \ 2 \ 0 \ 0 \ 0]$$

The result is:

$$y[n] = \dots + x[0]h[n] + x[1]h[n-1] + \dots$$

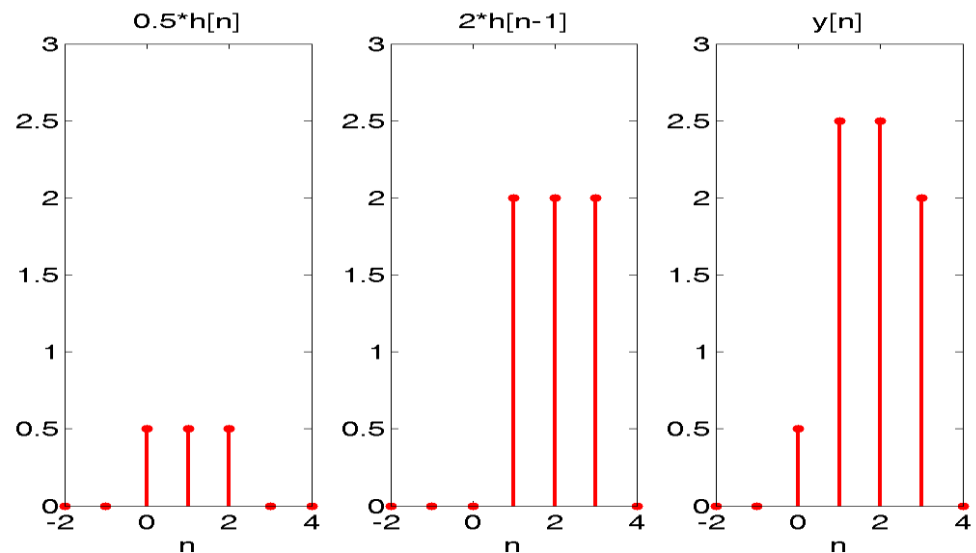
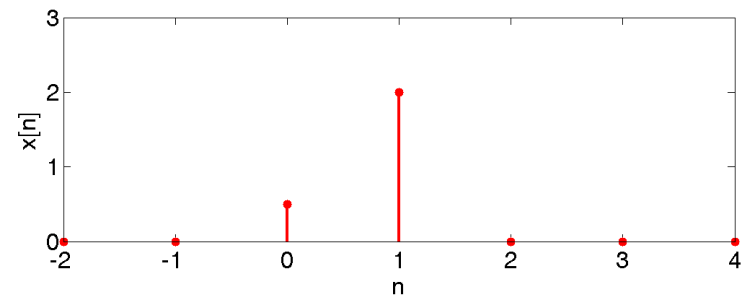
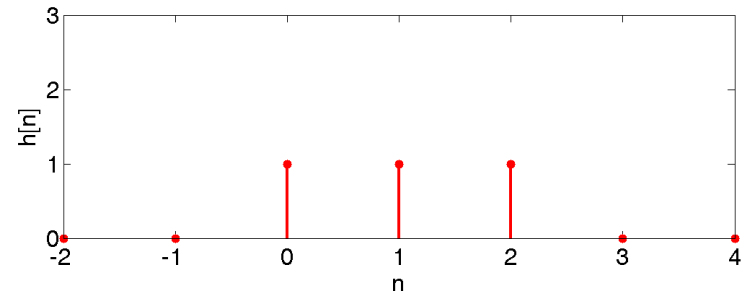
$$= 0 +$$

$$0.5*[0 \ 0 \ 1 \ 1 \ 1 \ 0 \ 0] +$$

$$2.0*[0 \ 0 \ 0 \ 1 \ 1 \ 1 \ 0] +$$

$$0$$

$$= [0 \ 0 \ 0.5 \ 2.5 \ 2.5 \ 2 \ 0]$$



Example 2: LTI Convolution

Consider the problem
described for example 1

Sketch $x[k]$ and $h[n-k]$ for any
particular value of n , then
multiply the two signals and
sum over all values of k .

For $n < 0$, we see that $x[k]h[n-k]$
 $= 0$ for all k , since the non-
zero values of the two
signals do not overlap.

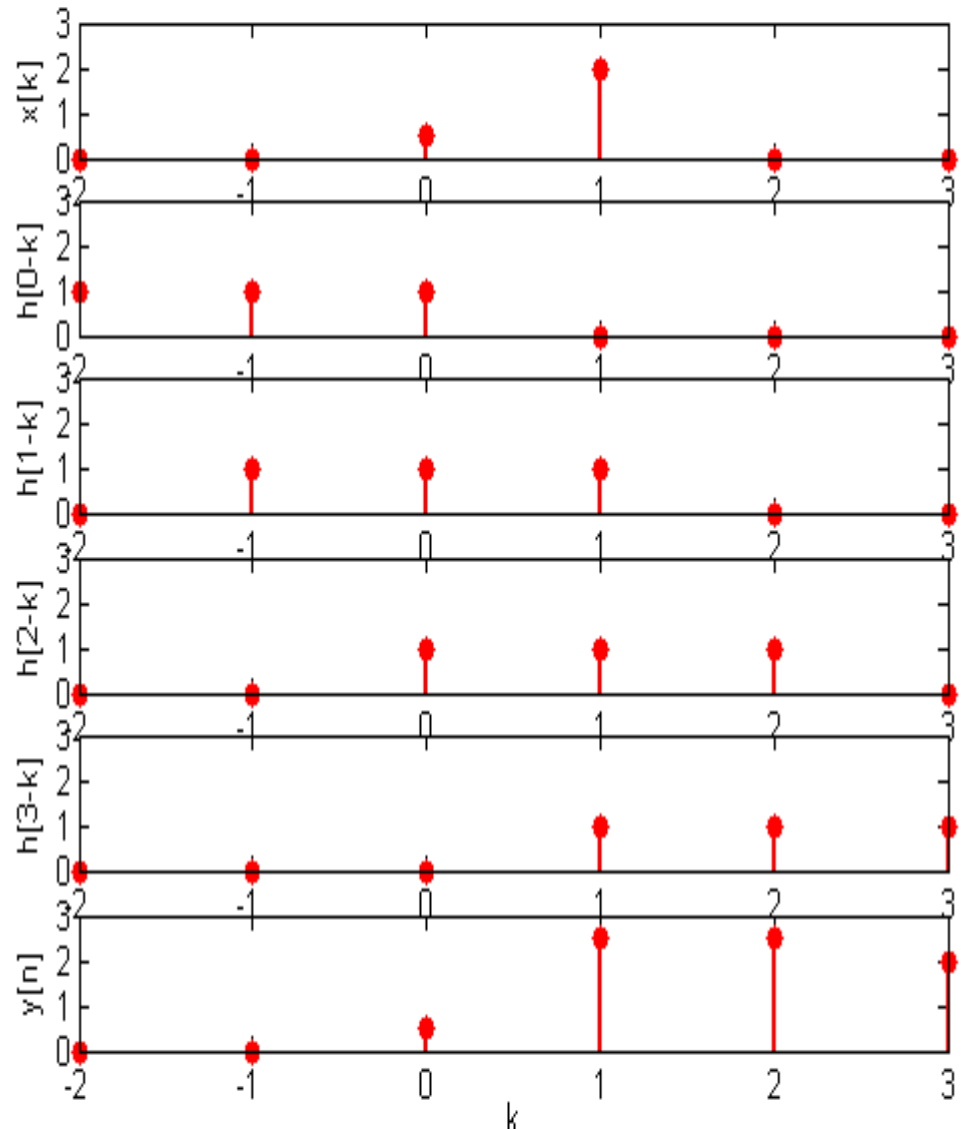
$$y[0] = \sum_k x[k]h[0-k] = 0.5$$

$$y[1] = \sum_k x[k]h[1-k] = 0.5 + 2$$

$$y[2] = \sum_k x[k]h[2-k] = 0.5 + 2$$

$$y[3] = \sum_k x[k]h[3-k] = 2$$

As found in Example 1



Example 3: LTI Convolution

Consider a LTI system that has a step response $h[n] = u[n]$ to the unit impulse input signal

What is the response when an input signal of the form

$$x[n] = \alpha^n u[n]$$

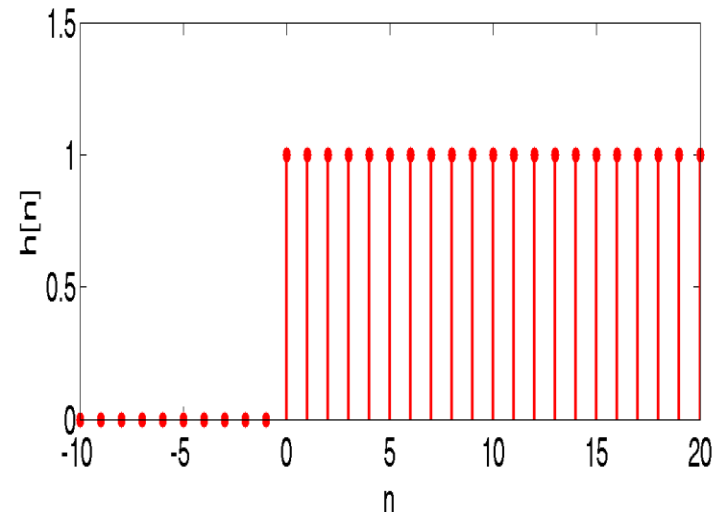
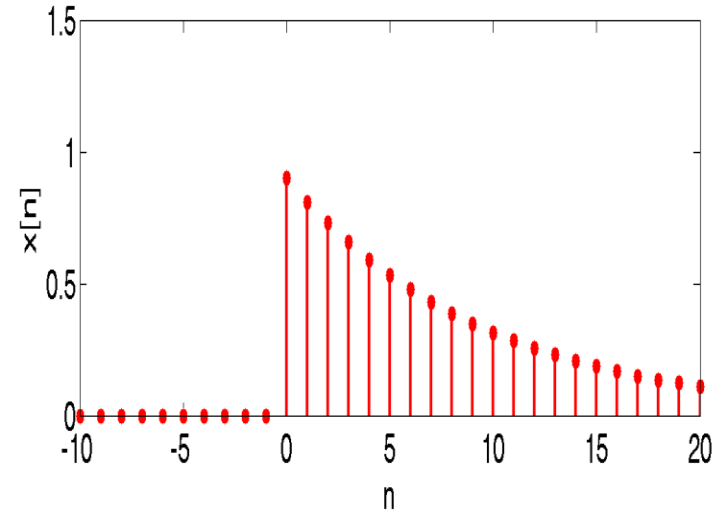
where $0 < \alpha < 1$, is applied?

For $n \geq 0$:

$$\begin{aligned} y[n] &= \sum_{k=0}^n \alpha^k \\ &= \frac{1 - \alpha^{n+1}}{1 - \alpha} \end{aligned}$$

Therefore,

$$y[n] = \left(\frac{1 - \alpha^{n+1}}{1 - \alpha} \right) u[n]$$



Discrete LTI Convolution in Matlab

In Matlab to find out about a command, you can search the help files or type:

```
>> lookfor convolution
```

at the Matlab command line. This returns all Matlab functions that contain the term “convolution” in the basic description

These include:

conv()

To see how this works and other functions that may be appropriate, type:

```
>> help conv
```

at the Matlab command line

Example:

```
>> h = [0 0 1 1 1 0 0];
```

```
>> x = [0 0 0.5 2 0 0 0];
```

```
>> y = conv(x, h)
```

```
>> y = [0 0 0 0 0.5 2.5 2.5 2 0 0 0 0 0]
```


Lecture 4: Summary

Any discrete LTI system can be completely determined by measuring its unit impulse response $h[n]$

This can be used to predict the response to an arbitrary input signal using the convolution operator:

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

The output signal $y[n]$ can be calculated by:

- Sum of scaled signals – example 1
- Non-zero elements of h – example 2

The two ways of calculating the convolution are equivalent

Calculated in Matlab using the `conv()` function (but note that there are some zero padding at start and end)

Lecture 4: Exercises

Q2.1-2.7, 2.21

Calculate the answer to Example 3 in Matlab, Slide 14